Project Report: High-Dimensional fMRI to Spatial 3D Object Reconstruction

Streamlit Live Deployment URL: https://fmriproject-dnp3anjwbw9fvht9z3mtix.streamlit.app/

1. Introduction & Project Scope

The main target of this project is to map brain activity recorded via functional Magnetic Resonance Imaging (fMRI) to the actual 3D spatial geometry of the physical object the human subject was viewing. This problem forms an interface between neuroimaging signal interpretation and 3D computer vision. The system translates high-dimensional biological time-series data into continuous spatial coordinates (X, Y, Z) representing 3D point cloud topologies.

2. Dataset Architecture & Preprocessing Pipeline

The source data structures contained a significant feature-to-target imbalance, which required specific dimensional alignments before model ingestion:

• **Input Voxel Tensor (X):** The initial shape was (533, 10, 31141). This represents 533 independent clinical scans recording brain voxel intensities across 31,141 specific locations across 10 temporal sequence snapshots.
• **Target Output Topology (obj_points):** The ground truth shape was (292, 2048, 3). This indicates 292 unique physical objects, where each object is defined by a dense array of 2,048 precise structural vertices.
• **Temporal Sequence Compression:** To scale down computational resource utilization, the time-series vector was compressed along the time axis using a temporal averaging operator (np.mean(axis=1)). This reduced the complex inputs into a continuous static 1D spatial feature map of dimension 31,141 for each subject sample.

3. Advanced Validation Strategy (Mitigating Data Leakage)

• **The Challenge:** Because only 292 unique 3D target shapes exist across 533 total scans, multiple fMRI records correspond to identical target objects. A standard random train-test division would allow the same object target to appear simultaneously in both sections, creating severe data leakage and artificially inflated scores.
• **The Solution:** A strict GroupKFold (5-Fold Cross-Validation) strategy was enforced using the dataset's native 'group' categorical array. By blocking target object IDs from overlapping across cross-validation splits, the network was forced to demonstrate true generalized inference capabilities on novel, unseen geometric shapes.

4. Deep Learning Architecture & Optimization Mechanics

A Multi-Layer Perceptron (MLP Baseline Architecture) was built inside PyTorch to approximate the mappings between neuro-voxel intensity levels and coordinate coordinates:

• **Layer Layout:** The network maps an input feature dimension of 31,141 onto a succession of hidden dense projections (256 units and 512 units) combined with non-linear ReLU activation functions before reaching the linear output surface.
• **Target Regression Flattening:** The target geometric coordinates were vectorized into a 1D sequence of length 6,144 (2048 points × 3 coordinate axes) to support continuous point regression.
• **Loss Metric & Hyperparameters:** Optimized via the Adam Optimizer (constant learning rate of 0.001) under a Mean Squared Error (MSE) loss criterion across 20 iterations using a mini-batch capacity of 32. The model smoothly converged to a final Validation MSE Loss of ~0.024.

5. Production Validation & Script Requirements

The compiled model outputs were run against the target directory checking script (check_submission.py). After handling minor configuration challenges, the pipeline passed with a complete 'Submission OK' verification stamp. The production directory tree is structured as follows:

```
submission/
■■■ predictions.npz           # (167, 2048, 3) predicted vertices + cleaned sample_id ke
ys
■■■ report.ipynb              # Replicable programmatic assignment documentation
■■■ metrics/
■   ■■■ val_metrics.json      # Certified evaluation scores (chamfer, composite, silhoue
tte)
■■■ ply/
■   ■■■ [sample_ids].ply      # 167 standalone ASCII point clouds formatted with pure x,
y,z rows
■■■ renders/
■   ■■■ [sample_ids]_view.png # 668 multi-view camera scene screenshots (256x256 frame s
pecs)
■■■ code/
    ■■■ workspace.ipynb       # Fully reproducible notebook pipeline
```

6. Streamlit Cloud Dashboard Deployment Architecture

To bring the model out of isolated research script files and into a clean, client-accessible system, the network pipeline was fully deployed on Streamlit Community Cloud.

• **Dynamic Array Handling:** The application ingests an unknown .npz archive directly from a drag-and-drop file uploader component, safely processing the numeric file vectors from the memory buffer.
• **Automated Data Normalization:** The input array automatically processes the fMRI feature matrices using a unified axis compression technique, making sure it perfectly aligns with the initial feature shapes.
• **Mathematical Topology Synthesis:** The prediction layer maps fMRI activation values directly to spatial coordinate deviations. This approach builds a valid 3D point array shape of exactly (2048, 3) nodes while preventing coordinate collapse.
• **Live Inspection Table:** Displays a raw interactive coordinate preview dataframe showcasing the immediate spatial distribution metrics.
• **Universal 3D Mesh Downloader:** Packages the generated results directly into a properly configured ASCII .PLY file format (including complete structural headers), allowing users to immediately download the 3D files for rendering in external software like Blender or MeshLab.

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